Computer Science

SYLLABUS

Grade: 12
Credit hrs: 5

Subject code: Com. 4281
Working hrs: 160

Scope and Sequence of Contents

S. N.	Content Area	Contents	Working Hour
1	Database Management System (DBMS)	1.1 Introduction to data, database, Database system, DBMS 1.2 Field, Record, Objects, Primary Key, Alternate key, Candidate key 1.3 Advantages of using DBMS 1.4 DDL (Data Definition Language) and DML (Data Manipulation Language) 1.5 Database Model: Network Model, Hierarchical Model, Relational database model 1.6 Concept of Normalization: 1NF, 2NF, 3NF 1.7 Centralized Vs. Distributed Database 1.8 Database Security	12
2	Data Communication and Networking	2.1 Basic elements of Communication System: 2.2 Concept of Communication System 2.3 Block Diagram of communication System /Model 2.4 Elements of Data Communication/Transmission 2.5 Simplex, Half duplex and Full duplex communication mode 2.6 Concept of LAN and WAN 2.7 Transmission Medium: Guided and Unguided 2.8 Transmission impairments terminology (Jitter, Singing, Echo, Crosstalk, Distortion, Noise, Bandwidth, Number of receivers) 2.9 Basic concept of Networks Architecture: Client- Server and Peer-to-peer 2.10 Some Basic Terms and Tool Used in Computer Network: IP Address, Sub Net Mask and Gateway, MAC address, Internet, Intranet, Extranet 2.11 Network Tool: Packet tracer, Remote Login 2.12 Network Connecting Devices: NIC, Modem, router, switch 2.13 Network Topologies: Bus, Ring and star topology 2.14 Basic Concept OSI Reference Model 2.15 Internet Protocol Addressing	15
3	Web Technology II	3.1 Introduction 3.2 Server side and Client Side Scripting 3.3 Introduction of internet technology 3.4 Adding Java script to HTML page 3.5 Java script fundamental 3.6 Java Script Data types 3.7 Variables and operators 3.8 Functions and control structure if-else, if-else- if, switch-case, for, while, do while loop 3.9 Object based programming with Java Script and Event handling 3.10 Image, event and form objects 3.11 Form validation, JQuery 3.12 Server Side Scripting using PHP 3.13 Introduction to PHP: Hardware and Software Requirements 3.14 Object oriented programming with server side scripting 3.15 Basic PHP syntax 3.16 PHP data types 3.17 Basic Programming in PHP 3.18 Operators (Arithmetic, logical, comparison, operator precedence) 3.19 Variables Manipulation 3.20 Database Connectivity	12

		3.21 Connecting server side script to database 3.22 Making SQL queries 3.23 Fetching data sets getting data about data 3.24 Creating SQL database with server side scripting 3.25 Displaying queries in tables	
4	Programming in C	4.1 Review of C programming concept 4.2 Functions • Concept of library and user defined functions and advantages • function definition, prototype, call and return statements • Accessing a Function by passing values • Concept of storage: automatic and external • Concept of Recursion: factorial and Fibonacci problems 4.3 Structures and Unions • Structure: Definition, Declaration, Initialization and Size of Structure. • Accessing member of structure • Array of structure • Union: Definition, Declaration • Difference between union and structure. 4.4 Pointers • Definition of Pointer • Address (&) and indirection (*) operator • Pointer Expression and Assignment • Call by values and call by reference 4.5 Working with Files • Concept of Data File • Sequential and Random File • File manipulation function: putw, getw, putc, getc, fscanf, fprintf • Opening, Reading, Writing and Appending data file	12
5	Object- Oriented Programming (OOP)	5.1 Programming paradigms: procedural, structural and object oriented 5.2 Features of OOP: Class, Object, Polymorphism and Inheritance 5.3 Advantages of OOP 5.4 Application of OOP	10
6	Software Process Model (SPM)	6.1 Software Project Concept 6.2 Concept of software development process 6.3 Concept SDLC life cycle 6.4 System Analyst Vs Software Engineer 6.5 Requirement Collection Methods 6.6 Concept of system design 6.7 Software and quality 6.8 Software development model: waterfall, prototype, agile	10
7	Recent Trends in Technology	7.1 Concept of Artificial Intelligence (AI) and Robotics 7.2 Concept of Cloud Computing 7.3 Concept of Big Data 7.4 Concept of Virtual Reality 7.5 Concept of e-com, e-medicine, e-gov. 7.6 Concept of Mobile Computing 7.7 Concept of Internet of things (IoT)	9
		Total	**80**

SPECIFICATION GRID

SN	Content Area	Working hour	Competency level																		Content Area /Unit wise Marks
			Remembering				Understanding						Applying						Higher Ability						
			MC Q		SA Q		MC Q		SA Q		LA Q		MC Q		SA Q		LA Q		MC Q		SA Q		LA Q		
			No. of Questions	Marks	No. of Questions	Marks	No. of Questions	Marks	No. of Questions	Marks	No. of Questions	Marks	No. of Questions	Marks	No. of Questions	Marks	No. of Questions	Marks	No. of Questions	Marks	No. of Questions	Marks	No. of Questions	Marks	
1	Database Management System (DBMS)	12																							8
2	Data communication and Networking	15																							9
3	Web Technology II	12	3	3	1	5	2	2	2	10	-	-	2	2	1	5	1	8	2	2	1	5	1	8	8
4	Programming in C	12																							8
5	Object-oriented Programming (OOP)	10																							6
6	Software Process Model (SPM)	10																							6
7	Recent Trends in Technology	9																							5
Total Marks		80	8				12						15						15						50

Item format plan					
S.N.	Type of item	Score per item	Total item	Total score	Time
1	Multiple Choice Questions	1	9	9	25 minutes
2	Short Question Answer	5	5	25	95 minutes
3	Long Question Answer	8	2	16	
Grand Total			16	50	2 hours

Remarks:

- Item format in composite should be met as per the specification grid.
- Designated weightage in the combined cell should be met, but ±3 marks variation will be allowed within a unit/content area. But no unit can be nil.
- In the case of SAQ and LAQ, these should ensure that 1 mark will be assigned per element expected as correct response.
- The distribution of cognitive domain of questions should be nearly 15% knowledge/remembering, 25% understanding, 30% applying and 30% higher ability level, but ±5 percent variation will be allowed in overall question set.
- SAQ and LAQ can be structured (have two or more sub-items). SAQ and LAQ can be distributed to two or more cognitive behaviors.
- In such case these will be added to their respective cognitive behavior. In sum the distribution of cognitive behavior should be approximately to the required distribution. In case of SAQ there will be 2 "OR" questions and in case of LAQ there will be 2 "OR" question.

CHAPTER BASED QUESTIONS

1. DATABASE MANAGEMENT SYSTEM (DBMS)

MULTIPLE CHOICE QUESTIONS [1 MARK]

1. 2080 Q.No. 2 Which of the following is DML statement?
   a. Create b. Select
   c. Create view Alter d. Drop
2. 2080 Q.No. 6 Which function is used to connect to mySQL database?
   a. mySqLi_open ( ) b. mySqLi_connect ( )
   c. var mySqLi_query ( ) d. $mySqLi_con ( )
3. 2079 GIE Set A Q.No. 1 Which statement is used to delete a table in SQL?
   a. REMOVE b. DROP
   c. Alter d. Rename
4. 2079 GIE Set A Q.No. 2 With SQL, how do you select all the records from a table named "NEB" where the value of the column "department" is "CDC"?
   a. Select * from NEB where department is exactly CDC;
   b. Select * from NEB where department is equal to 'CDC';
   c. Select * from NEB where department='CDC';
   d. Select * from NEB where department is not in CDC;
5. 2079 GIE Set B Q.No. 1 Which key is an attribute of a table that matches a primary key attribute in another table to create a relationship?
   a. Candidate b. Primary
   c. Foreign d. National
6. 2079 GIE Set B Q.No. 2 What is called the user has identified via a username and password in DBMS?
   a. Authentication b. Authorization
   c. Accounting d. Auditing
7. 2079 Set A Q.No. 1 Which of the following SQL statement is used to DELETE rows from a database table?
   a. DELETE b. REMOVE
   c. DROP d. CLEAR
8. 2079 Set A Q.No. 2 A field that is used to uniquely define a particular record in a table is called:
   a. Primary Key b. Entity
   c. Relationship d. Constraints
9. . . . is not the data type of MS-Access.
   a. Number b. Text
   c. Memo d. Form
10. . . . is the DBMS.
    a. Fox-Pro b. MS Excel
    c. Lotus-123 d. All of the above
11. The primary key does not accept . . .
    a. Text b. Number
    c. Null Value d. None of above
12. In MS Access, we enter data using . . .
    a. Form b. Table
    c. Report d. Both a and b
13. Date/Time occupies . . . bytes of memory.
    a. 4 b. 2
    c. 8 d. 16

Answers									
1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
b	b	b	c	c	a	a	a	d	a
11.	12.	13.							
c	d	c							

SHORT ANSWER QUESTIONS [5 MARKS]

1. 2080 Q.No. 10 Explain the Relational Data model with example. [5]
2. 2080 Q.No. 10 OR Write SQL DDL commands to execute the following task with reference to the schema given below :
   student_info (regno as integer, name as character (25), class integer, gender character (1), address character (25).
3. 2080 Q.No. 11 Define a syntax for database connectivity. Write a server side scripting code to insert data into the table student having fields (first name, last name, mark and email:) Assume that
   server name ="localhost", username ="root", password =" ", database name ="student DB". [1+4]
4. 2079 GIE Set A Q.No. 10 Explain Relational database model. [5]
5. 2079 GIE Set A Q.No. 10 OR Demonstrate any two DDL statements with an example. [5]
6. 2079 GIE Set B Q.No. 10 What are the roles of the database administrator? Describe. [5]
7. 2079 GIE Set B Q.No. 10 OR Describe the DDL statement with an example in SQL. [5]
8. 2079 Set A Q.No. 10 Differentiate the centralized and distributed database system. [5]
9. 2079 Set A Q.No. 10 OR What are the purposes of normalization? Give an example of 2NF. [5]
10. 2078 Set C Q.No. 8 Explain the different database model with suitable examples. [5]
11. 2078 Set C Q.No. 9 Explain entity, attribute and relationship in Relational Database Management System. [5]
12. 2078 Set C Q.No. 10 Explain the role of Database Administrator (DBA). [5]
13. 2077 Set D Q.No. 4 Describe 1NF, 2NF and 3NF normalization concept to database design. [5]
14. 2076 GIE Set A Q.No. 7 What are the importances of normalization in DBMS? Explain. [5]
15. 2076 GIE Set A Q.No. 8 Explain E-R diagram in DBMS. [5]
16. 2076 GIE Set B Q.No. 7 Describe the normalization in DBMS? [5]
17. 2076 GIE Set B Q.No. 8 What are the major functions of DBA? Explain. [5]
18. 2076 GIE Set B Q.No. 10 2075 Set A Q.No. 8 Differentiate between centralized and distributed database. [5]
19. 2076 Q.No. 7 Explain the different types of data integrity in DBMS. [5]

20. 2076 Q.No. 8 What is SQL? Describe the DML with an example. [2+3]
21. 2075 GIE Q.No. 8 What is database model? Explain relational model. [2+3]
22. 2075 Set B Q.No. 8 Explain the advantages of Database Management System. [5]
23. 2075 Set B Q.No. 11 Explain the benefits of decentralized database models. [5]
24. 2074 Supp Q.No. 7 What is Relational Database Management System? Explain. [5]
25. 2074 Supp Q.No. 8 Explain DML and DDL. [2.5+2.5]
26. 2074 Set A Q.No. 7 Explain the data integrity with example. [5]
27. 2074 Set A Q.No. 8 Describe the normalization in database. [5]
28. 2074 Set B Q.No. 7 What are the role of DBA? Explain. [5]
29. 2074 Set B Q.No. 8 What are 1NF and 2NF normalization principle in database? Explain. [5]
30. 2073 Supp Q.No. 8 What is data security? Explain. [5]
31. 2073 Supp Q.No. 14 What is normalization? Explain first and second normal form with example. [1+4]
32. 2073 Set C Q.No. 7 What is DBMS? Explain data integrity with suitable example. [2+3]
33. 2073 Set C Q.No. 8 Define DML and DDL with example. [3+2]
34. 2073 Set D Q.No. 8 What is DBMS? Differentiate between centralized and distributed database. [2+3]
35. 2073 Set D Q.No. 9 Explain relational database model. [5]
36. 2072 Set C Q.No. 7 Define DBMS. List out the objectives of DBMS. [1+4]
37. 2072 Set C Q.No. 8 What is relational database model? List the advantages and disadvantages of relational database model. [1+4]
38. 2072 Set C Q.No. 9 Describe the terms 'SQL' and 'DML'.[2.5+2.5]
39. 2072 Set D Q.No. 10 Describe any two database model with diagram. [5]
40. 2072 Set E Q.No. 7 What is normalization? Write the advantages of normalization. [2+3]
41. 2072 Set E Q.No. 9 2069 Q.No. 9 What is E-R diagram? Explain the advantages of E-R diagram in system design.[1+4]
42. 2072 Set E Q.No. 14 OR What are the advantages of RDBMS. [5]
43. 2071 Supp Q.No. 10 Who is database administrator? List the roles of database administrator. [1+4]
44. 2071 Set C Q.No. 7 What is database? List out the advantages of Database Management System. [1+4]
45. 2071 Set C Q.No. 9 Describe the centralized and distributed database systems. [5]
46. 2071 Set D Q.No. 8 What is RDBMS? List out the function of RDBMS. [1+4]
47. 2070 Supp Q.No. 7 Explain the database models with clear diagrams. [5]
48. 2070 Supp Q.No. 8 What is DBMS? List out the functions of DBMS. [1+4]
49. 2070 Set C Q.No. 7 What is Hierarchical database model? List out the advantages and disadvantages of Hierarchical database model. [1+4]
50. 2070 Set C Q.No. 9 Explain the centralized and distributed database models. [2.5+2.5]
51. 2070 Set D Q.No. 8 Differentiate between DBMS and RDBMS with examples. [2.5+2.5]
52. 2069 Q.No. 8 What is database? List the major uses of database application software. [1+4]
53. 2069 Q.No. 10 Explain the database model with suitable examples. [5]
54. 2068 Cancelled Q.No. 7 2057 Q.No. 3 Define the term RDBMS. State the guiding principles used in database design. [2+5]
55. 2068 Cancelled Q.No. 9 What is data integrity? State and describe different types of data integrity. [2+5]
56. 2068 Cancelled Q.No. 11 What is normalization? How it is used in data analysis? Explain. [2+5]
57. 2068 Q.No. 4 What do you understand by the term data integrity? Why it is important in designing a database? [2+5]
58. 2068 Q.No. 9 Explain the benefits of centralized database. [7]
59. 2067 Q.No. 8 2065 Q.No. 6 2063 Q.No. 8 What is normalization? Explain the normalization process with examples. [2+5]
60. 2067 Q.No. 11 What do you understand by the term data integrity? Why it is important thing to be considered while designing a database? Explain [2+5]
61. 2066 Supp Q.No. 10 What is the purpose of E-R diagram? List out the different symbols used in E-R diagram with their meanings and examples. [7]
62. 2066 Q.No. 7 Define database and DBMS. Explain the advantages of Database system over flat-file system. [2+5]
63. 2066 Q.No. 10 What are the advantages of distributed database system over centralized database system? [7]
64. 2066 Supp Q.No. 6 What is data model? Explain about different data modeling techniques. [2+5]
65. 2065 Q.No. 10 2061 Q.No. 6 What is data security? How it can be implemented? [2+5]
66. 2064 Q.No. 4 Differentiate between database and database management system. Explain the top down methodology of database design. [3+4]
67. 2064 Q.No. 8 Explain the benefits of centralized database. What are the major responsibilities of a Database Administrator? Explain in detail. [3+4]
68. 2064 Q.No. 11 What do you mean by data security? Explain briefly the type of security methods normally adopted in computerized environment. [2+5]
69. 2063 Q.No. 6 Define database administrator. Explain the duties and responsibilities of a database administrator.
70. 2061 Q.No. 4 Who is database administrator? Explain the benefits of centralized database system. [3+4]
71. 2060 Q.No. 7 Give two advantages of using a database rather than a number of separate files. What are major responsibilities of a database administrator? [3+4]

72. 2060 Q.No. 9 Why is normalization needed? Explain normalization process with examples. [2+5]
73. 2059 Q.No. 4 Differentiate between Database and DBMS. Explain the top down methodology of Database design. [3+4]
74. 2059 Q.No. 11 Explain the benefits of Centralized Database. What are major responsibilities of a Database Administrator? What are the objectives of system analysis? [2+3+2]
75. 2058 Q.No. 2 Name the different models of DBMS. Write short notes on any three of the following: (a) SQL (b) Data Dictionary (c) ERD (d) DDL [1+6]
76. 2058 Q.No. 3 What are Domains and Tuples? What are the characteristics of a good data model? [2+5]
77. 2058 Q.No. 8 What is prototyping? Write short notes on any two of the following: (a) Normalization of data (b) Data security (c) RDBMS [2+5]
78. 2058 Q.No. 9 What is the role of Database Administrator? What are the duties and responsibilities of a System Analyst? [2+5]

LONG ANSWER QUESTIONS [8 MARKS]

79. 2078 Set C Q.No. 4 Define normalization. Explain 2NF and 3NF with example. [4+6]
80. 2075 GIE Q.No. 5 What is normalization? Write its advantages. Explain second and third normal form with example. [2+2+6]
81. 2075 Set A Q.No. 5 What is normalization? Explain first, second and third normal form with example. [2+8]
82. 2075 Set B Q.No. 5 What are the importance of normalization? Explain second and third normal form with example. [4+6]
83. 2062 Q.No. 2 OR What do you understand by the term data integrity? Why it is an important thing to be considered while designing a database? State and describe different types of data integrity. [20]
84. 2059 Q.No. 2
    a. What do you understand by the term data integrity? Why is it important thing to be considered while designing a database? State and describe different types of data integrity. [5]
    b. What is data security? How it can be implemented? [5]

WRITE SHORT NOTES ON

85. 2076 GIE Set A Q.No. 15b 2074 Set B Q.No. 15a 2071 Supp Q.No. 15(ii) Data Security [2.5]
86. 2076 Q.No. 15a Primary Key [2.5]
87. 2074 Set A Q.No. 15a ER diagram [2.5]
88. 2073 Set C Q.No. 15a DBMS [2.5]
89. 2073 Set D Q.No. 15b 2071 Set C Q.No. 15b 2070 Set D Q.No. 15a 2064 Q.No. 12b 2062 Q.No. 12c Normalization [2.5]
90. 2070 Set C Q.No. 15b 2068 Q.No. 12a 2057 Q.No. 7a Data dictionary [2.5]
91. 2073 Supp Q.No. 15b 2059 Q.No. 5(a) DML [3.5]
92. 2057 Q.No. 4(b) Advantage of DBMS [3.5]
93. 2057 Q.No. 4(c) Data modeling [3.5]

2. DATA COMMUNICATION AND NETWORKING

MULTIPLE CHOICE QUESTIONS [1 MARK]

1. 2080 Q.No. 3 A collection of hyperlinked documents on the Internet is called :
   a. HTML b. webpage
   c. www d. email system
2. 2080 Q.No. 4 Which of following IPV4 address valid?
   a. 192.168.1.1 b. 257.2.5.1
   c. EEE.000.33.000 d. 202.70.71.262
3. 2080 Q.No. 9 Coaxial cable is an example of ...
   a. Communication software
   b. Guided Transmission media
   c. Unguided Transmission media
   d. All of them
4. 2079 GIE Set A Q.No. 3 In which topology connected with other two computers in circular form?
   a. Star b. Bus
   c. Ring d. Hybrid
5. 2079 GIE Set B Q.No. 3 Which is default subnet mask for the class C IP address?
   a. 255.0.0.0 b. 255.255.255.0
   c. 255.255.0.0 d. 255.255.255.255
6. 2079 Set A Q.No. 3 What does "MAC" stands for in MAC Address?
   a. Mandatory Access Control b. Media Access Control
   c. Micro Access Control d. Media Access Certificate
7. . . . refers to the sending, receiving, and processing of information by electronic means.
   a. computer b. communication
   c. software d. Program
8. Which refers to the communication media?
   a. UTP cable b. Fiber optic cable
   c. Satellite d. All of the above
9. Which one is not network topology?
   a. Star b. Ring
   c. Client server d. BUS
10. Which is not a guided media?
    a. UTP Cable b. STP Cable
    c. Satellite d. Fiber Optic
11. Which device is necessary to operate the Internet?
    a. CD-ROM b. ISP
    c. Modem d. All of the above

Answers									
1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
c	a	b	c	b	b	b	d	c	c
11.									
c									

SHORT ANSWER QUESTIONS [5 MARKS]

1. 2079 GIE Set B Q.No. 11 Describe the class C IP address with an example. [5]

2. 2079 GIE Set B Q.No. 11 OR Compare the star and ring topology with its features. [5]
3. 2078 Set C Q.No. 11 Explain about coaxial cable and fiber optics cable. [5]
4. 2077 Set D Q.No. 5 Compare the star and ring topology. [5]
5. 2076 GIE Set A Q.No. 9 What is computer network? List out hardware equipments to establish the computer network.[2+3]
6. 2076 GIE Set A Q.No. 10 2074 Set B Q.No. 10 Differentiate between LAN and WAN. [5]
7. 2076 GIE Set B Q.No. 9 Describe the different types of communication media in the Networking. [5]
8. 2076 Q.No. 9 Define network topology. Explain the Ring Topology. [2+3]
9. 2076 Q.No. 10 Describe the wireless network system. List out devices and equipments necessary for Wi-Fi network. [3+2]
10. 2075 GIE Q.No. 9 What is topology? Explain Ring topology. [2+3]
11. 2075 GIE Q.No. 10 Differentiate between bounded and unbounded transmission media. [5]
12. 2075 Set A Q.No. 9 Differentiate between peer-to-peer and client/server network. [5]
13. 2075 Set A Q.No. 10 What is transmission media? Write advantages of optical fiber cable. [1+4]
14. 2075 Set B Q.No. 10 What is transmission media? Explain any two different types of transmission media. [1+2+2]
15. 2074 Supp Q.No. 9 What are the functions of router and switch? [2.5+ 2.5]
16. 2074 Set A Q.No. 9 Describe the Star topology with suitable diagram. [5]
17. 2074 Set A Q.No. 10 What is optical fiber cable in networking system? [5]
18. 2074 Set B Q.No. 9 Explain Unshielded Twisted Pair (UTP) cable and Coaxial cable. [2.5+2.5]
19. 2073 Supp Q.No. 9 Explain peer-to-peer and client/server network. [2.5+2.5]
20. 2073 Set C Q.No. 9 Describe the Bus and Star topology with suitable diagram. [2.5+2.5]
21. 2073 Set C Q.No. 10 Differentiate between guided and unguided networking media. [5]
22. 2073 Set D Q.No. 10 Explain Bus and Star topology with advantages and disadvantages. [2.5+2.5]
23. 2072 Set C Q.No. 10 Define computer network and explain its uses. [2+3]
24. 2072 Set C Q.No. 11 Describe 'simplex', 'half duplex', and 'full duplex' with example. [1+2+2]
25. 2072 Set D Q.No. 9 2070 Set C Q.No. 8 What is Networking? List the advantages of Networking. [1+4]
26. 2072 Set D Q.No. 11 Describe Network topologies with diagram. [5]
27. 2072 Set E Q.No. 8 What is computer network? Write the advantages and disadvantages of computer network. [1+4]
28. 2071 Supp Q.No. 7 What is computer network? Explain its uses. [2+3]
29. 2071 Set C Q.No. 8 What is Network topology? Describe any two network topologies with clear diagram. [1+4]
30. 2071 Set C Q.No. 10 Describe the 'Coaxial cable' and 'Satellite' with examples. [2.5+2.5]
31. 2071 Set D Q.No. 10 What is Network? List out the benefits of Networks. [1+4]
32. 2070 Supp Q.No. 9 Define computer networks and explain their uses. [1+ 4]
33. 2070 Supp Q.No. 10 Describe 'Simplex', 'half duplex' and 'full duplex' channel of communications with examples. [5]
34. 2070 Set C Q.No. 10 Explain any two transmission media with appropriate diagrams. [2.5+2.5]
35. 2070 Set D Q.No. 10 Describe the types of network topologies with clear diagrams. [5]
36. 2069 Q.No. 7 What are the types of LAN topologies? Explain with diagrams. [5]
37. 2068 Cancelled Q.No. 6 2065 Q.No. 4 2062 Q.No. 6 What is local area network? Explain different types of topologies with diagrams. [2+5]
38. 2068 Q.No. 10 What are the types of LAN topology? Explain with diagram. [7]
39. 2067 Q.No. 6 What is network topology? Explain the different types of network topologies with diagram. [7]
40. 2066 Q.No. 4 Define Network Architecture. Explain client-server and peer-to-peer networking. [2+5]
41. 2066 Q.No. 6 Explain about OSI/ ISO model of networking. [7]
42. 2066 Supp Q.No. 5 What is the purpose of transmission media? Explain its types. [2+5]
43. 2064 Q.No. 10 What is networking? Distinguish between star topology and ring topology of networking principles with the help of clean diagram. [2+5]
44. 2063 Q.No. 10 What is networking? Explain the advantages and disadvantages of networking. [2+5]
45. 2061 Q.No. 7 What are the different types of LAN topology? Explain. [7]
46. 2060 Q.No. 5 What is network topology? What are the different types of LAN topology? Explain. [3+4]
47. 2060 Q.No. 10 What do you mean by transmission media? Write short notes on UTP cable and fibre optical cable.[3+2+2]
48. 2058 Q.No. 4 What do you mean by topology? What do you understand by the following terms (any two): (a) Client-Server Network (b) Workstation (c) Protocol [1+6]
49. 2057 Q.No. 5 What is a local area network? List the different types of topologies with diagrams. [2+5]

LONG ANSWER QUESTIONS [8 MARKS]

50. 2080 Q.No. 15 What is network topology? Differentiate between guided (wire media) and unguided (wireless media) with example. [2+6]
51. 2079 GIE Set A Q.No. 15 Explain client-server and peer -to-peer network architecture with diagram.
52. 2079 Set A Q.No. 15 Compare the star and ring topology with pros and cons. Which of data communication cable is more appropriate to design the local area network? Describe. [5+3]

WRITE SHORT NOTES ON

53. 2076 GIE Set B Q.No. 15b 2075 GIE Q.No. 15b Router [2.5]
54. 2076 Q.No. 15b IP address [2.5]
55. 2075 Set A Q.No. 15 2073 Set D Q.No. 15a WiFi [2.5]
56. 2074 Supp Q.No. 15a LAN [2.5]
57. 2074 Set A Q.No. 15b 2073 Supp Q.No. 15a TCP/IP [2.5]
58. 2074 Set B Q.No. 15b Bluetooth [2.5]
59. 2073 Set C Q.No. 15b 2066 Supp Q.No. 12b Network protocols [2.5]
60. 2072 Set E Q.No. 15a 2069 Q.No. 15b 2067 Q.No. 12 b 2062 Q.No. 12b Satellite [2.5]
61. 2071 Supp Q.No. 15(i) Protocol [2.5]
62. 2069 Q.No. 15a 2067 Q.No. 12 a 2062 Q.No. 12(a) Coaxial Cable [2.5]
63. 2068 Cancelled Q.No. 12 b Packet switching [2.5]
64. 2063 Q.No. 7 (i) Client server network (ii) Workstation (iii) Protocol [5]
65. 2061 Q.No. 8(b) Fiber-optic Cable [5]
66. 2059 Q.No. 5(b) Bus-Topology [5]
67. 2059 Q.No. 5(c) Transmission media [5]

3. WEB TECHNOLOGY II

MULTIPLE CHOICE QUESTIONS [1 MARK]

1. 2080 Q.No. 1 Which of the given sign is used as a shortcut for JQuery?
   a. the % sign b. the & sign
   c. the $ sign d. the @ sign
2. 2080 Q.No. 5 Which JQuery method is used to hide selected elements?
   a. hide ( ) b. hide (on)
   c. invisible ( ) d. display (none)
3. 2079 GIE Set A Q.No. 4 Which method is used to returns today's date and time in PHP?
   a. Today date () method.
   b. Today's date () method.
   c. Date () method.
   d. Today's date and time () method.
4. 2079 GIE Set A Q.No. 5 What is the correct syntax for referring to an external JavaScript script?
   a. <script src="neb.js"></script>
   b. <Java href="neb"></j s>
   c. <Script href-="neb.js"></script>
   d. Js script src="neb.js"></script>
5. 2079 GIE Set B Q.No. 4 What would be the value of "z" in the given code in JavaScript?
   var x = 5; var y = 2; var z = x % y;
   a. 1 b. 2
   c. 0 d. 2.5
6. 2079 GIE Set B Q.No. 5 Which is the correct open tag and close tag for PHP scripts?
   a. {} b. <?php?>
   c. <> d. <??>
7. 2079 Set A Q.No. 4 What is the correct syntax for referring to an external JavaScript script?
   a. <script src="myscript.js"></script>
   b. <script href="myscript.js"></script>
   c. <js href="myscript.js"></js>
   d. <js src="myscript.js"></js>
8. 2079 Set A Q.No. 5 Which of the following is the correct way of defining a variable in PHP?
   a. $variable name = value; b. $variable_name = value;
   c. $variable_name = value d. $variable name as value;
9. Java script is . . .
   a. procedural oriented language
   b. object oriented programming language
   c. Firmware
   d. None of the above
10. Data types in java script are . . .
    a. primary and secondary b. primitive and non-primitive
    c. local and universal d. None of the above
11. PHP stands for . . .
    a. Hypertext Pre-processor b. program hyper processor
    c. Process hyper program d. None of the above
12. concat() is . . .
    a. integer method b. string method
    c. image method d. None of the above
13. . . . is an identifier whose value may change during the execution of the program.
    a. constant b. function
    c. variable d. None of the above

Answers									
1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
c	a	c	a	a	b	a	b	b	b
11.	12.	13.							
a	b	c							

SHORT ANSWER QUESTIONS [5 MARKS]

1. 2080 Q.No. 11OR Write a JavaScript code to calculate the factorial of a given number. [5]
2. 2079 GIE Set A Q.No. 11 Write a function to enter two numbers and find sum of two numbers in JavaScript.
3. 2079 GIE Set A Q.No. 11 Or Explain the database connection method in PHP.
4. 2079 Set A Q.No. 11 Write a program to find the factorial of any given number using Javascript. [5]
5. 2079 Set A Q.No. 11 OR How do you fetch data from database in PHP and display it in form? Describe. [5]
6. Write short notes on Internet technology.
7. Differentiate between server-side and client-side scripting.
8. Explain different data types used in JavaScript.
9. What is JQuery? Write down its uses.
10. What is PHP? Explain its uses and advantages.
11. Write down the basis PHP syntax and an example of a PHP program.

12. Explain the different operators used in a PHP.
13. Write a PHP program to display the largest among three numbers.

LONG ANSWER QUESTIONS [8 MARKS]

14. 2079 GIE Set B Q.No. 15 What are the uses of Java Script in web page development? How do you connect the PHP and My SQL database? Explain with connection class. [3+2]
15. What is Object-based programming? Explain different JavaScript objects.
16. Write down the server-side script to create a database, connect with it, create a table and insert data in it.

4. PROGRAMMING IN C

MULTIPLE CHOICE QUESTIONS [1 MARK]

1. 2080 Q.No. 7 Which of the following is true about FILE *fpt?
   a. FILE is a keyword in c for representing files and fpt is a variable of FILE type
   b. FILE is a buffered Stream.
   c. FILE is a Stream.
   d. FILE is a structure and fpt is a pointer to the structure of FILE type.
2. 2079 GIE Set A Q.No. 6 What is the output of this statement "printf(%d", (a++)) in C programming?
   a. The current value of a.
   b. The value of a increment by 1.
   c. Error
   d. Garbage
3. 2079 GIE Set B Q.No. 6 Which statement(s) is/are true to open a binary file "Record.dat" for reading and writing?
   a. file = open("Record.dat","wb")
   b. file = open("Record.dat","ab")
   c. file = open("Record.dat","rb")
   d. file = open("Record.dat","rb+")
4. 2079 Set A Q.No. 6 What is the output of C program?

```
void main(){ int b=25;
//b memory location=1234;
int *p; p=&b; printf("%d %d", &b, p);}
```

   a. 25 25 b. 1234 1234
   c. 25 1234 d. 1234 25
5. . . . . is a reserved word that has standard and predefined meanings in C.
   a. variable b. identifier
   c. keyword d. None of the above
6. . . . is a quantity that doesn't change during program execution
   a. variable b. constant
   c. data type d. None of the above
7. . . . is a symbolic name, which is used to store different types of data in the computer's memory and may change during the program execution.
   a. variable b. constant
   c. identifier d. None of the above
8. . . . is a symbol that instructs C to perform some operation or action on one or more operands.
   a. operand b. operation
   c. operator d. None of the above
9. . . . is something that an operator acts on.
   a. operand b. operation
   c. operator d. None of the above

Answers									
1.	2.	3.	4.	5.	6.	7.	8.	9.	
d	c	a	a	c	b	a	c	a	

SHORT ANSWER QUESTIONS [5 MARKS]

1. 2078 Set C Q.No. 12 Define the term "constant", "identifier" and "keyword" in C program. [5]
2. 2077 Set D Q.No. 6 What is variable? List the major data types which is used in C program. [2+3]
3. 2076 GIE Set A Q.No. 11 What is structure in C Program? Explain. [5]
4. 2076 GIE Set B Q.No. 11 List out the different types of operator in C. Describe the logical operator with example.[2+3]
5. 2076 GIE Set B Q.No. 12 Define pointer in C with example.[5]
6. 2076 Q.No. 12 2074Set A Q.No. 12 2073 Set C Q.No. 12 Describe the different mode of the file handling concept in C.[2.5+2.5]
7. 2074 Supp Q.No. 11 2074 Set B Q.No. 11 Explain the file handling concept in C. [2+3]
8. 2072 Set C Q.No. 12 2069 Q.No. 8 Or 2061 Q.No. 10 Differentiate between array and structure with example. [5]
9. 2071 Supp Q.No. 9 Explain the array and structure with examples. [2.5+2.5]

LONG ANSWER QUESTIONS [8 MARKS]

10. 2080 Q.No. 16 Write the advantage of pointer. Write a C-program to enter the radius of a football and find the area of football by using user defined function. [3+5]
11. 2080 Q.No. 16OR Define the structure. Write a C-program using structure to input staff id, name and the salary of 50 staffs. Display staff id, name and salary of those staff whose salary range from 25 thousand to 40 thousand. [2+6]
12. 2079 GIE Set A Q.No. 16 Write a C program to enter name, grade, age and address of 10 students in a structure and display the information.
13. 2079 GIE Set A Q.No. 16 Or Describe file handling concept in C. Write a C program to enter names and address of the student and store them in a data file "student.dat".
14. 2079 GIE Set B Q.No. 16 Define function. Write a program to generate factorial of given number using recursive function in C program. [3+5]
15. 2079 GIE Set B Q.No. 16 OR Explain the three modes of file opening in C. Write a program to reads name and roll from a file named "exam.dat" and display them. [5+3]
16. 2079 Set A Q.No. 16 What is structure? Write a program to input roll, name and age of 5 students and display them properly using structure. [2+6]
17. 2079 Set A Q.No. 16 OR Write a C program to enter ID, employee_name, and post of the employee and store them

in a data file named "emp.txt". Display each record on the screen in an appropriate format. [8]

18. 2078 Set C Q.No. 1a Explain any three different types of operators in C with example. [5]
19. 2078 Set C Q.No. 1b Write a program to print fibonacci series of 6, 12, 18, 30 ... upto 10 terms using loop. [5]
20. 2078 Set C Q.No. 2a What is pointer? Write the advantages of pointer. [1+4]
21. 2078 Set C Q.No. 2b What is string function? Write a program to concatenate two strings. [2+3]
22. 2078 Set C Q.No. 5 Write a program to input emp_no, name and salary of 100 employees using structure. Display the name and salary of employee using structure. [10]
23. 2077 Set D Q.No. 3a Describe the pointer with example. [4]
24. 2076 GIE Set A Q.No. 3 How do you define function in C program? Write a program to find out given number is negative or positive using function. [4+6]
25. 2076 GIE Set A Q.No. 5 What is file handling concept in C? Write a program to enter name and salary of employee and write it in a file "employee.txt" [2+8]
26. 2076 GIE Set B Q.No. 2 Write a program to read the marks of any 5 students in a subject and count how many students are passed or failed. (Pass marks = 35) [10]
27. 2076 GIE Set B Q.No. 3 Describe the function in C. Write a program to find out factorial of given number using function. [4+6]
28. 2076 GIE Set B Q.No. 5 Explain the structure in C. Write a program to enter the 5 student name, age and sex using structure and display. [3+7]
29. 2076 Q.No. 1 What is user defined function? Write a program to input length and breadth of a pond and find its area by using user defined function. [4+6]
30. 2076 Q.No. 4 What is structure? Write a program to input employee id, name, address and post of 20 employees and display them properly. [2+8]
31. 2076 Q.No. 5a What is the importance of file pointer in file handling? Explain with an example. [5]
32. 2075 GIE Q.No. 3 What is function? Write the advantages of function. Differentiate between library and user-defined function with example. [1+3+6]
33. 2075 GIE Q.No. 4 Write a program to enter name, post and age of 10 employee's using structure and display in proper format. [10]
34. 2075 Set A Q.No. 2 Write a program to input 10 integer numbers into an array and display the sum of the numbers. [10]
35. 2075 Set A Q.No. 3 What is function? Explain any four string functions with example. [2+8]
36. 2075 Set A Q.No. 4 Write a program to input name, roll, and reg. no. of 10 student's using structure and display in proper format. [10]
37. 2075 Set B Q.No. 3 What is function? Write a program to read the principle, time and rate. Find the simple interest using function. [4+6]
38. 2075 Set B Q.No. 4 Write a program to store 50 records of students using fields, std. no., name and mark. Display the name of student and mark. [10]
39. 2074 Supp Q.No. 3 What are the importance of function in C program? Write a program to find out given number is odd or even using function. [4+6]
40. 2074 Supp Q.No. 4 What are the string function in C? Write a program to print the given text in reverse order. [4+6]
41. 2074 Supp Q.No. 5 Describe the structure in C. Write a program to enter the 5 subject marks and calculate sum and print using structure. [2+8]
42. 2074 Set A Q.No. 4 Describe the function in C. Write a program to find the sum of n integer number using function. [4+6]
43. 2074 Set A Q.No. 5 Write a program to input the 5 employee's name and salary using structure and display the records in proper format. [10]
44. 2074 Set B Q.No. 3 What are the methods of function declaration and call in C? Write a program to find out given number is odd or even using function. [4+6]
45. 2074 Set B Q.No. 5 What is structure in C? Write a program to enter the 20 employee's name, age and salary using structure and print them. [2+8]
46. 2073 Supp Q.No. 1b Differentiate between array and structure. [5]
47. 2073 Supp Q.No. 3 Define control structure. Write a program to find the sum, difference and greater of two integer numbers using switch case statement. [3+7]
48. 2073 Supp Q.No. 4a Write a program to multiply two integer numbers using functions [5]
49. 2073 Supp Q.No. 5a What is pointer? Define address of ($) and indirection (*) operator. [5]
50. 2073 Supp Q.No. 5b Write a program to enter name, address and salary of a employee and store it in a file "staff.dat".
51. 2073 Set C Q.No. 4 Describe the function in C. Write a program to find the sum of 'n' integer number using function in C. [4+6]
52. 2073 Set C Q.No. 5 Differentiate between structure and union. Write a program to input the employee name and post of five employees and display the records in proper format using structure. [3+7]
53. 2073 Set D Q.No. 4a Write a program to add two integer numbers using function. [5]
54. 2073 Set D Q.No. 5a What is pointer? Write its advantages. [5]
55. 2073 Set D Q.No. 5a Write a program to enter name, post and salary of a employee and write it in a file "employee.dat"
56. 2072 Set C Q.No. 4a What is function? List out the advantages of functions. [1+4]
57. 2072 Set C Q.No. 5
    a. Describe fscanf and fprintf function. [5]
    b. Write a program which asks name, age, roll-number of student and write it in a file student.dat. [5]

58. 2072 Set D Q.No. 4a Differentiate between Structure and Union. [2.5+2.5]

59. 2072 Set D Q.No. 5
   a. Describe any two file handling functions. [2.5+2.5]
   b. Write a program to display name, age and address reading from file "record.dat". [5]

60. 2072 Set E Q.No. 3 What is a function? Write a function to add two integer number. [3+7]

61. 2072 Set E Q.No. 5 2069 Q.No. 5 Write a program which reads name, department and age from a file named "employee. dat" and display them on the monitor. [10]

62. 2071 Supp Q.No. 5 What is "fscanf" function? Write a program to display name, age and address reading from a file named "record.dat". [2+8]

63. 2071 Set C Q.No. 1 What is function? Write a program to generate factorial of a given number using recursive function.[2+8]

64. 2071 Set C Q.No. 2 Describe any five 'file handling function' with examples. [10]

65. 2071 Set D Q.No. 3 What is pointer? Describe the benefits of pointer with examples. [4+6]

66. 2071 Set D Q.No. 5 Write a program which reads name, roll-number and age from a file named "student.dat" and display them. [10]

67. 2070 Supp Q.No. 2 Describe the types of loop with flowchart and examples. [10]

68. 2070 Supp Q.No. 3 Write a program which asks the user to input 'n' terms of number and find out the greatest and smallest number among those numbers. [10]

69. 2070 Supp Q.No. 4 Differentiate between array and structure with suitable examples. [5 + 5]

70. 2070 Supp Q.No. 5 Write a program which asks name, age, roll number of students and write them in a file "xyz.dat". [10]

71. 2070 Set C Q.No. 2 Describe any five "string handling functions" with examples. [10]

72. 2070 Set C Q.No. 3 Describe array, structure and pointer with examples. [10]

73. 2070 Set C Q.No. 5 Describe fprintf and fscanf file handling functions. Write a program which writes "welcome to Nepal" in a file. [10]

74. 2070 Set D Q.No. 3 Differentiate between structure and union with suitable examples. [5+5]

75. 2070 Set D Q.No. 4 What is recursion? Write a program to calculate factorial value of given number using recursive function. [2+8]

76. 2070 Set D Q.No. 5 Write a program which reads name, department and age from a file named "employee.dat" and display them. [10]

77. 2068 Q.No. 1d What do you mean by "local", "global" and "static" variables? Explain with examples. [5]

78. 2068 Q.No. 2a What is an array? Write down the similarities and differences of array with pointer. [2+3+5]

79. 2068 Q.No. 3
   a. Write program to sort an array of 'n' elements in descending order. [10]
   b. Write a program to enter "name", "roll-no" and "marks" of 10 students and store them in file. Read and display the same from the file. [10]

80. 2068 Cancelled Q.No. 2
   a. What is recursive function? Write a program to calculate the factorial of an integer using recursion. [3+7]
   b. Define pointer. Discuss the relation between the pointer and array with suitable examples. [2+8]

81. 2068 Cancelled Q.No. 3a Write a program to open a new file and read roll-no, name address and phone number of students until the user says "no", after reading the data, write it to the file then display the content of the file. [10]

82. 2067 Q.No. 1b Write a recursive function to calculate the factorial of any integer number. [5]

83. 2067 Q.No. 2
   a. How do you declare an array? Write a program to arrange the elements of an array in ascending order.[3+7]
   b. What is the difference between library function and user-defined function? Write a program using user-defined function to calculate y raise to power x. [5+5]

84. 2067 Q.No. 3
   a. Differentiate between array and structure with suitable example. [10]
   b. Write a program to delete and rename the data file using remove and rename command. [10]

85. 2066 Supp Q.No. 3
   a. Write a program to show data writing and reading operation to/from a data file. [2+8]
   b. Explain the importance of pointer in C-Programming. Illustrate pointer operations with examples. Show the relationship between array and pointers with example.[3+3+4]

86. 2066 Q.No. 2
   a. Write a program to store name and mark of 20 students. Sort the data according with mark in descending order and display them. [10]
   b. Write a program to find the sum of 'n' integer numbers using function. [10]

87. 2066 Q.No. 3
   a. Write a program to store std-no, name and mark of n' students in a data file. Display the records in appropriate format reading from the file. [10]
   b. Differentiate array and structure with examples. [10]

88. 2065 Q.No. 2b Write differences between structure and union with syntax. [10]

89. 2065 Q.No. 3a Write a program to enter name, roll-number and marks of 10 students and store them in the file. [10]

90. 2064 Q.No. 1c Write a program for reading a data file. [5]

91. 2064 Q.No. 3a Write a program to delete and rename data file using remove and rename command. [10]

92. 2063 Q.No. 1b Write a C program to calculate the factorial of a given number using functions.

93. 2063 Q.No. 3
   a. Differentiate between structure and pointer with examples. [10]
   b. Write a C program design a menu based system which has the following features: [10]
      (i) Appending record (ii) Reading record
      (iii) Delete record (iv) Quit

94. 2062 Q.No. 3b Write a C program that will read successive records from the new data file and display each record on the screen in an appropriate format. [10]

95. 2061 Q.No. 1b Write a program that reads different names and addresses into the computer and rearrange the names into alphabetical order using the structure variables. [10]

96. 2061 Q.No. 3
   a. Write a program using C language that reads successive records from the new data file and display each record on the screen in an appropriate format. [10]
   b. What is pointer? Explain the meaning of each of the following declarations: [2.5+7.5]
      (i) int *p; (ii) int *p[10]; (iii) int (*) [10]
      (iv) int *p [void]; (v) int * P(char *a)

5. OBJECT-ORIENTED PROGRAMMING (OOP)

MULTIPLE CHOICE QUESTIONS [1 MARK]

1. 2079 GIE Set A Q.No. 7 Which of the following is not a OOPs concept?
   a. Class b. Exception.
   c. Polymorph. d. Encapsulation

2. 2079 GIE Set B Q.No. 7 What is called in class to wrapping /hiding of data into a single unit?
   a. Abstraction b. Polymorphism
   c. Encapsulation d. Inheritance

3. 2079 Set A Q.No. 7 Which feature of OOP is illustrated the code reusability?
   a. Polymorphism b. Abstraction
   c. Encapsulation d. Inheritance

4. OOP stands for . . .
   a. Operator overloading programming.
   b. Object-Oriented Programming.
   c. Object overloading programming.
   d. None of the above.

5. . . . is a concept or thing with defined boundaries that are relevant to the problem we are dealing with.
   a. object b. class
   c. variable d. None of the above

6. . . . is a collection or group of similar objects that have the same properties, common behavior, and relationships.
   a. object b. class
   c. constant d. None of the above

7. . . . is a property that allows the reuse of an existing class to build a new class.
   a. class b. operation
   c. inheritance d. Object

8. . . . is the process that allows selective hiding of data and functions in a class.
   a. class b. inheritance
   c. encapsulation d. Polymorphism

Answers									
1.	2.	3.	4.	5.	6.	7.	8.		
b	c	d	b	a	b	c	c		

SHORT ANSWER QUESTIONS [5 MARKS]

1. 2080 Q.No. 12 How is even-driven program (or OOP) differing from procedural - oriented programming? Explain. [5]
2. 2079 GIE Set A Q.No. 12 Write the advantages of OOPs. [5]
3. 2079 GIE Set B Q.No. 12 What is inheritance? Explain with an example. [5]
4. 2079 Set A Q.No. 12 Compare the OOPs and procedural programing language. [5]
5. 2075 GIE Q.No. 11 Distinguish between OOP and procedural language. [5]
6. 2075 Set A Q.No. 11 2071 Set D Q.No. 11 What are the importance of OOP? [5]
7. 2073 Supp Q.No. 10 What is OOP? Write the application of OOP. [1+4]
8. 2073 Set D Q.No. 11 Explain class and object. [2.5+2.5]
9. 2072 Set D Q.No. 12 What is OOP? List the characteristics of OOP. [1+4]
10. 2071 Supp Q.No. 14 2067 Q.No. 9 2063 Q.No. 5 2061 Q.No. 11 Explain the terms polymorphism and inheritance. [5]
11. 2071 Set C Q.No. 12 2070 Supp Q.No. 12 What is OOP? List the advantages of OOP. [1+4=5]
12. 2070 Set C Q.No. 12 What is procedural oriented programming? Explain. [5]
13. 2070 Set D Q.No. 11 2075 GIE Q.No. 1b Explain polymorphism and inheritance with examples. [2.5+2.5]
14. 2069 Q.No. 14 Write the advantages and disadvantages of OOP [5]
15. 2068 Cancelled Q.No. 5 2057 Q.No. 8 What is object-oriented programming? How is it different from the procedure oriented programming? [3+4]
16. 2068 Q.No. 6 2062 Q.No. 8 Why polymorphism and inheritance are important concepts of OOP? Explain. [7]
17. 2066 Supp Q.No. 11 Explain the characteristics of objects oriented programming. [7]
18. 2058 Q.No. 5 Name the tools that are used in program design. Polymorphism and Inheritance are important concepts of OOP. What d you understand by there two terms?
19. 2058 Q.No. 6 What is a Logical operator? Give two examples of their use. Compared to procedure-oriented programming what advantage does not events driven programming offer? [3+4]

LONG ANSWER QUESTIONS [8 MARKS]

20. 2078 Set C Q.No. 3b 2075 Set B Q.No. 2a Write the importance of OOPs. [5]
21. 2076 GIE Set A Q.No. 1a 2075 Set A Q.No. 1b Define the terms class and objects. [2.5]

WRITE SHORT NOTES ON:

22. 2078 Set C Q.No. 15b 2063 Q.No. 6c Polymorphism [2.5]
23. 2075 Set B Q.No. 15 2066 Q.No. 12a Inheritance [5]

6. SOFTWARE PROCESS MODEL

MULTIPLE CHOICE QUESTIONS [1 MARK]

1. 2080 Q.No. 8 Which a SDLC phase is used to ensure quality software?
   a. implementation b. maintenance
   c. testing d. system study
2. 2079 GIE Set A Q.No. 9 In which phase of SDLC software code is done?
   a. Analysis b. Design
   c. Development. d. Maintenance.
3. 2079 GIE Set B Q.No. 8 What software development approach should a company use if they want to consider the input of users in the development process?
   a. RAD b. Agile
   c. Waterfall d. Prototype
4. 2079 Set A Q.No. 8 Which of following is the discovering requirement from a user in the requirement collection process?
   a. Feasibility study b. Requirement Elicitation
   c. Requirement Specification d. Requirement validation
5. A set of programs written for a computer to perform a particular task is called
   a. hardware b. software
   c. Both a and b d. None of the above
6. . . . defines a sequence of tasks that must be carried out to build new software.
   a. computer
   b. Hardware
   c. Software development process
   d. None of the above
7. SDLC stands for . . .
   a. Scheme Development Life Cycle
   b. Software/System Development Life Cycle
   c. Software/System Design Life Cycle
   d. Scheme Design Life Cycle
8. . . . is also called a Preliminary Investigation or Need Analysis Phase.
   a. System design b. System study
   c. System development d. System testing
9. . . . is to determine whether the whole process of system analysis leading to computerization would be worthwhile for the organization or not.
   a. Feasibility study b. System design
   c. System development d. Testing

Answers									
1.	2.	3.	4.	5.	6.	7.	8.	9.	
c	c	b	b	b	c	b	b	a	

SHORT ANSWER QUESTIONS [5 MARKS]

1. 2080 Q.No. 13 Explain the importance of the system testing of the system development life cycle (SDLC). [5]
2. 2079 GIE Set A Q.No. 13 Explain requirement analysis phase of SDLC.
3. 2079 GIE Set B Q.No. 13 Describe the waterfall software development model with pros and cons. [5]
4. 2079 Set A Q.No. 13 What are the major activities performed to design the software? Describe. [5]
5. 2078 Set C Q.No. 6 What is feasibility study? Explain. [5]
6. 2078 Set C Q.No. 7 Describe the desirable characteristics of a system analyst. [5]
7. 2076 GIE Set A Q.No. 6 What are roles of system analyst in SDLC phase? [5]
8. 2076 GIE Set B Q.No. 6 What are the importances of SDLC in software development process? [5]
9. 2076 Q.No. 6 Define SDLC. Describe the feasibility analysis methods. [2+3]
10. 2075 GIE Q.No. 6 Explain system analysis phase of SDLC. [5]
11. 2075 GIE Q.No. 7 2075 Set A Q.No. 7 2073 Supp Q.No. 7 Define the terms DFD and E-R diagram. [2.5+2.5]
12. 2075 Set A Q.No. 6 2073 Set D Q.No. 6 2065 Q.No. 5 Who is system analyst? Explain the role of system analyst. [1+4]
13. 2075 Set B Q.No. 6 List out different stages of system development life cycle and explain any two stages of them. [1+2+2]
14. 2075 Set B Q.No. 7 Describe the major role and responsibility of system analyst in SDLC phase. [5]
15. 2075 Set B Q.No. 9 What is E-R diagram? Write the meaning of graphical symbols used in E-R diagram. [5]
16. 2074 Supp Q.No. 6 List the steps of SDLC. Explain System Analysis Phase. [2+3]
17. 2074 Set A Q.No. 6 Describe the waterfall model to explain the SDLC. [5]
18. 2074 Set B Q.No. 6 What are the software development process phases in SDLC? Explain any one phase. [2+3]
19. 2073 Supp Q.No. 6 Explain the system development phase.[5]
20. 2073 Set C Q.No. 6 What are the major activities in SDLC with based on waterfall model? [5]
21. 2073 Set D Q.No. 7 Write the importance and necessity of SDLC. [5]
22. 2072 Set C Q.No. 6 Describe SDLC with diagram. [5]
23. 2072 Set D Q.No. 6 Describe the components of feasibility study. [5]
24. 2072 Set D Q.No. 7 Who is system analyst? List out characteristics of system analyst. [1+4]
25. 2072 Set D Q.No. 8 Describe system flowchart with diagram. [5]

26. 2072 Set E Q.No. 6 2070 Set C Q.No. 6 2068 Q.No. 5 2067 Q.No. 5 2064 Q.No. 9 2062 Q.No. 5 What is feasibility study? Explain different levels of feasibility study. [2+3]
27. 2072 Set E Q.No. 13 Write the importance of SDLC. [5]
28. 2071 Supp Q.No. 6 What is feasibility study? Why it is necessary before designing a system? [2+3=5]
29. 2071 Supp Q.No. 11 OR What is expert system? Explain the fields of expert system. [1+4]
30. 2071 Set C Q.No. 6 What is system analysis and design? Describe briefly. [5]
31. 2071 Set D Q.No. 6 Describe different levels of feasibility study. [5]
32. 2071 Set D Q.No. 7 2070 Set D Q.No. 7 2062 Q.No. 4 Who is system analyst? List out the roles of system analyst. [1+4]
33. 2070 Supp Q.No. 6 Explain SDLC with appropriate diagram.[5]
34. 2070 Set D Q.No. 6 What is system? Explain the basic elements of system. [1+4]
35. 2069 Q.No. 6 Explain different stages of system development life cycle with clear figure. [5]
36. 2068 Cancelled Q.No. 4 What is feasibility study? Why feasibility study is important in system analysis phase? Explain. [2+5]
37. 2068 Cancelled Q.No. 8 2057 Q.No. 2 Who is system analyst? Highlight the characteristics of a good analyst. [2+5]
38. 2068 Cancelled Q.No. 10 What are the documentation techniques? Explain with an example. [7]
39. 2068 Q.No. 7 Define program logic. Explain different types of program logic tools. [2+5]
40. 2068 Q.No. 8 What is system analysis? What are the major objectives of system analysis? Explain. [2+5]
41. 2067 Q.No. 4 What is system analyst? Explain the major roles of system analyst. [2+5]
42. 2066 Q.No. 5 Explain about the system design methodologies. [7]
43. 2066 Q.No. 9 What are the different program logic tools? Explain about the decision table and decision tree with examples. [2+5]
44. 2066 Q.No. 11 Explain about the different testing techniques during the system development. [7]
45. 2066 Supp Q.No. 4 How can you test a newly developed system? Explain different types of testing techniques. [2+5]
46. 2066 Supp Q.No. 8 Differentiate between decision table and decision tree with example. [7]
47. 2065 Q.No. 11 What is documentation? Explain the importance of documentation in program designing. [2+5]
48. 2064 Q.No. 6 Who is system analyst? Explain the different stages of system development life cycle. [2+5]
49. 2063 Q.No. 4 What is system analysis? Explain the different steps of system development life cycle. [2+5]
50. 2063 Q.No. 11 What are the different symbols used to construct a flow chart? Give brief explanation along with a neat diagram. [2+5]
51. 2062 Q.No. 11 What is program logic? Explain the different types of program logic design tools. [2+5]
52. 2061 Q.No. 5 Define feasibility study. Why feasibility study is important in system development process. Explain. [2+5]
53. 2061 Q.No. 9 What are the documentation techniques? Explain any three of them. [2.5+4.5]
54. 2060 Q.No. 4 What do you mean by feasible study? Why is it necessary before designing a system? [4+3]
55. 2060 Q.No. 8 What do you mean by program algorithm? Explain the aims of program design and building blocks of a structured programming [3+4]
56. 2059 Q.No. 9 State the steps involved in system development cycle in chronological order. Explain data security laws. [5+2]
57. 2058 Q.No. 1 What is the importance of Data flow diagrams in the S.D.L.C? What do you mean by the following terms (any two): [3+4]
    a. Physical system b. C.A.S.E
    c. P.I.E.C.E.S d. Feasibility Study
58. 2057 Q.No. 1 Define the term system. List different stages of system development life cycle. Highlight the importance of the first two stages. [2+3+2]

WRITE SHORT NOTES ON

59. 2076 GIE Set B Q.No. 15a Feasibility analysis [2.5]
60. 2072 Set D Q.No. 15a 2071 Set D Q.No. 15b 2070 Set D Q.No. 15 b Expert System [2.5]
61. 2067 Q.No. 7a 2063 Q.No. 12 (ii) 2057 Q.No. 4(a) System security. [5]
62. 2065 Q.No. 12 b 2063 Q.No. 12 (i) Feasibility study [3.5]

7. RECENT TRENDS IN TECHNOLOGY

MULTIPLE CHOICE QUESTIONS [1 MARK]

1. 2079 GIE Set A Q.No. 8 IoT stands for . . .
    a. International of Technology
    b. Internet of Things.
    c. Internet of Technology
    d. International of Things.
2. 2079 GIE Set B Q.No. 9 Which is not types of cloud services
    a. SaaS b. PaaS
    c. IaaS d. CaaS
3. 2079 Set A Q.No. 9 What devices are detecting and responding to changes in an environment that is embedded in smart phones and an integral part of the Internet of Things (IoT)?
    a. Wi-Fi b. Barcode
    c. RFID d. Sensors

4. . . . is the part of computer science concerned with designing intelligent computer systems.
   a. Hardware b. Computer
   c. Artificial Intelligence d. None of the above
5. . . . is the branch of technology that deals with the design, construction, operation, and application of robots.
   a. Hardware b. Robotics
   c. Software d. None of the above
6. . . . is the use of various services, such as software development platforms, servers, storage, and software, over the Internet.
   a. Hard disk b. Cloud computing
   c. Pen drive d. None of the above
7. SaaS stands for . . .
   a. Software as a service b. System as a service
   c. Both a and b d. None of the above
8. IaaS stands for . . .
   a. Internet as a service b. Infrastructure as a service
   c. Both a and b d. None of the above

Answers									
1.	2.	3.	4.	5.	6.	7.	8.		
b	d	d	c	b	b	a	b		

SHORT ANSWER QUESTIONS [5 MARKS]

1. 2080 Q.No. 14 What is cloud computing? Point out the advantage and disadvantage of cloud computing. [1+4]
2. 2079 GIE Set A Q.No. 14 Explain the concept of cloud computing.
3. 2079 GIE Set B Q.No. 14 Write a short note on e-governance and e-commerce. [3+2]
4. 2079 Set A Q.No. 14 Explain the popular five-application areas of AI. [5]
5. 2078 Set C Q.No. 13 How can you minimize computer crime? Explain. [5]
6. 2078 Set C Q.No. 14 What is E-commerce? List out demerit of E-commerce. [1+4]
7. 2077 Set D Q.No. 7 What is AI? Describe the applications of AI. [2+3]
8. 2076 GIE Set B Q.No. 13 What are the importance of e-learning in the 21th century? Write the importance of multimedia in e-learning. [3+2]
9. 2076 Q.No. 14 What are the roles of AI in our real life? List out AI related systems in our society. [1+4]
10. 2075 GIE Q.No. 14 Define the terms e-governance and e-medicine. [2.5+2.5]
11. 2075 Set A Q.No. 12 2073 Supp Q.No. 13 2072 Set E Q.No. 12 OR Define the terms e-business and e-learning. [2.5+2.5]
12. 2074 Supp Q.No. 13 Explain Artificial Intelligence. [2+3]
13. 2074 Set B Q.No. 13 What are the advantages and disadvantages of social media? List out. [5]
14. 2075 Set B Q.No. 14 What is E-commerce? Point out the advantages of E-commerce. [1+4]
15. 2073 Supp Q.No. 11 What is AI? Write the social impact of ICT. [2+3]
16. 2073 Set D Q.No. 12 What is AI? Explain cyber law. [2+3]
17. 2073 Set D Q.No. 14 Define the terms e-learning and virtual reality. [2.5+2.5]
18. 2072 Set C Q.No. 13 What is E-governance? List out the objectives of E-governance. [1+4]
19. 2072 Set D Q.No. 14 Describe applications of AI. [5]
20. 2072 Set E Q.No. 12 What is AI? What are the uses of AI? [2+3]
21. 2071 Supp Q.No. 11 What is e-governance? List the importance of e-governance. [1+4]
22. 2071 Set C Q.No. 13 Describe the application areas of AI. [5]
23. 2071 Set D Q.No. 14 Describe the objectives of e-governance. [5]
24. 2070 Supp Q.No. 14 2070 Set C Q.No. 13 2070 Set C Q.No. 13 2069 Q.No. 11 OR What is AI? Explain the application areas of AI. [1+4]
25. 2070 Supp Q.No. 15 List out the advantages and disadvantages of e-business. [5]
26. 2070 Set D Q.No. 14 What are the objectives of e-governance? Explain. [5]
27. 2069 Q.No. 11 What are the key challenges of implementing e-governance in developing countries? [5]
28. 2068 Q.No. 11 2061 Q.No. 12 What do you understand by AI? How it affects the modern society? [2+5]
29. 2066 Supp Q.No. 7 What is e-commerce? Explain its role to enhance the digital economy. [2+5]
30. 2064 Q.No. 5 2059 Q.No. 8 What do you understand by AI? How it affects the society? Explain the terms polymorphism and inheritance in terms of OOPs. [1+2+4]
31. 2060 Q.No. 11 What is e-commerce? Write impact of e-commerce technology in our society. [3+4]
32. 2058 Q.No. 7 What do you mean by 'e-commerce'? Distinguish between WWW (World Wide Web) and Internet? What is multimedia? What are the hardware devices required by a personal computer to make it 'multimedia capable'? [2+2+1+2]
33. 2057 Q.No. 7 Who does an AI application work? Can a computer really think with the aid of AI?

WRITE SHORT NOTES ON

34. 2078 Set C Q.No. 15a Use of AI [2.5]
35. 2075 GIE Q.No. 15a 2075 Set A Q.No. 15 AI [2.5]
36. 2075 Set B Q.No. 15 2060 Q.No. 6b Application of AI [5]
37. 2074 Supp Q.No. 15b Social Media [2.5]
38. 2072 Set D Q.No. 15b 2070 Set C Q.No. 15 a E-learning. [2.5]
39. 2071 Set C Q.No. 15 a 2067 Q.No. 7b 2066 Q.No. 12a 2063 Q.No. 12iii
    E-commerce [2.5]

EXAM QUESTION

Exam 2080

Sub. Code: 4281

NEB-Grade XII
2080 (2023)
Computer Science
New Course
(For regular students only)

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Time: 2 hrs **Full Marks: 50**

Group 'A'

Very short answer questions. **[9x1=9]**

1. Which of the given sign is used as a shortcut for JQuery?
   a. the % sign b. the & sign
   c. the $ sign d. the @ sign
2. Which of the following is DML statement?
   a. Create b. Select
   c. Create view Alter d. Drop
3. A collection of hyperlinked documents on the Internet is called :
   a. HTML b. webpage
   c. www d. email system
4. Which of following IPV4 address valid?
   a. 192.168.1.1 b. 257.2.5.1
   c. EEE.000.33.000 d. 202.70.71.262
5. Which JQuery method is used to hide selected elements?
   a. hide ( ) b. hide (on)
   c. invisible ( ) d. display (none)
6. Which function is used to connect to mySQL database?
   a. mySqLi_open ( ) b. mySqLi_connect ( )
   c. var mySqLi_query ( ) d. $mySqLi_con ( )
7. Which of the following is true about FILE *fpt?
   a. FILE is a keyword in c for representing files and fpt is a variable of FILE type
   b. FILE is a buffered Stream.
   c. FILE is a Stream.
   d. FILE is a structure and fpt is a pointer to the structure of FILE type.
8. Which a SDLC phase is used to ensure quality software?
   a. implementation b. maintenance
   c. testing d. system study
9. Coaxial cable is an example of ...
   a. Communication software
   b. Guided Transmission media
   c. Unguided Transmission media
   d. All of them

Answers								
1.	2.	3.	4.	5.	6.	7.	8.	9.
c	b	c	a	a	b	d	c	b

Group 'B'

Short answer questions **[5x5=25]**

10. Explain the Relational Data model with example. [5]
    OR
    Write SQL DDL commands to execute the following task with reference to the schema given below :
    student_info (regno as integer, name as character (25) , class integer, gender character (1), address character (25).
11. Define a syntax for database connectivity. Write a server side scripting code to insert data into the table student having fields (first name, last name, mark and email:) Assume that
    server name ="localhost", username ="root", password =" ", database name ="student DB". [1+4]
    OR
    Write a JavaScript code to calculate the factorial of a given number. [5]
12. How is even-driven program (or OOP) differing from procedural - oriented programming? Explain. [5]
13. Explain the importance of the system testing of the system development life cycle (SDLC). [5]
14. What is cloud computing? Point out the advantage and disadvantage of cloud computing. [1+4]

Group 'C'

Long answer questions **[8×2=16]**

15. What is network topology? Differentiate between guided (wire media) and unguided (wireless media) with example. [2+5]
16. Write the advantage of pointer. Write a C-program to enter the radius of a football and find the area of football by using user defined function. [3+5]
    OR
    Define the structure. Write a C-program using structure to input staff id, name and the salary of 50 staffs. Display staff id, name and salary of those staff whose salary range from 25 thousand to 40 thousand. [2+6]

MODEL QUESTIONS

Grade: Grade XII **Subject: Computer Science (Th)**
Full Marks: 50 (9 marks Obj + 41 Marks Sub) **Time: 2 Hours**

Issued by NEB, 2079

Group A

Rewrite the correct options of each question in your answer sheet. **[9×1=9]**

1. In which normal form of database, atomicity is introduced?
   a. First b. Second
   c. Third d. Fourth
2. Which of the following techniques is not implemented to protect a database?
   a. Rollback b. Backup
   c. Recovery d. Firewall
3. Which one of the following SQL commands is executed to display all the records having a second letter in the LNAME (LAST NAME) as "A" from the customer table?
   a. SELECT * FROM CUSTOMER WHERE LNAME LIKE "?A%";

b. SELECT * FROM CUSTOMER WHERE LNAME LIKE "_A%";
c. SELECT * FROM CUSTOMER WHERE LNAME LIKE "A%";
d. SELECT * FROM CUSTOMER WHERE LNAME LIKE "%A";

4. Which of the following is an incorrect IP address?
   a. 192.168.0.1 b. 192.168.1
   c. 172.255.0.0 d. 202.10.79.4
5. Which of the following is a server-side scripting language?
   A. JavaScript b. MySql
   c. PHP d. Jquery
6. Which of the following keywords are used to declare a variable in JavaScript?
   a. int or var b. float or let
   c. var or let d. char or var
7. Which of the following commands is executed in PHP to concatenate the variables $x with $y?
   a. $x + $y b. $x=$y
   c. concat ($x,$y) d. $x.$y
8. Which statement is incorrect about the object-oriented approach?
   a. Emphasis is on data rather than procedure.
   b. Data is hidden and cannot be accessed.
   c. Objects communicate through functions.
   d. It supports abstract data but not the class.
9. Which of the following feasibility study is concerned with cost benefit analysis?
   a. Technical feasibility b. Economic feasibility
   c. Operational feasibility d. Schedule feasibility

Answers									
1.	2.	3.	4.	5.	6.	7.	8.	9.	
a	d	b	b	c	c	d	d	b	

Group B

Short Answer Questions [5×5=25]

10. Which type of database system (centralized or distributed) is mostly preferred by financial institution like a bank? Give any four suitable reasons. [1+4]
    OR
    Nowadays most of the business organizations prefer applying relational model for database design in comparison to other models: Justify the statement with your arguments.[5]
11. Develop a program in JavaScript to exchange/swap the values of any two variables. [5]
    OR
    How can you connect a database with PHP? Demonstrate with an example. [5]
12. Describe the concept of Object Oriented and Procedure Oriented Programming in brief. [2+3]
13. Write down any five qualities of good software. [5]
14. Explain mobile computing with its advantages and disadvantages. [5]

Group C

Long Answer Questions

Give the long answers to the following questions. [2×8=16]

15. Why do most of the business organizations prefer setting their network with the client-server architecture? Write its advantages and disadvantages. [2+6]
16. Develop a program in C using structure to ask the information of any 12 students with roll_number, name and marks scored in sub1, sub2, and sub3. Also, display them in proper format along with the calculation of total and percentage. [Note: the full marks of each subject is 100].
    OR
    Demonstrate a program in C to create a data file named score.dat to store students' information with Reg_no, name, gender, and address. The program should ask the user to continue or not. When finished, the program should also display all the records in the proper format.

Issued by CDC, 2078

Group A

Multiple Choice questions [9×1=9]
Time: 20 minutes
Tick the correct answer.

1. Which of the statements are used in DDL?
   a. Create, alter and drop b. Create, insert and select
   c. Insert, update and delete d. Delete, alter and drop
2. With SQL, how do you select all the records from a table named "Persons" where the value of the column "FirstName" ends with an "a"?
   a. SELECT * FROM Persons WHERE FirstName="a"
   b. SELECT * FROM Persons WHERE FirstName LIKE "a%"
   c. SELECT * FROM Persons WHERE FirstName LIKE "%a"
   d. SELECT * FROM Persons WHERE FirstName="%a%"
3. Which of the following statements is true about a star network topology?
   a. Each device is connected to a switch or hub
   b. Each device is connected to each other
   c. Each device is connected in a trunk
   d. Each device is connected to a terminal
4. Which of the following is the correct syntax to display "Stay Safe" in an alert box using JavaScript?
   a. alert-box("Stay Safe"); b. confirm("Stay Safe");
   c. msgbox("Stay safe"); d. alert("Stay Safe");
5. What is the use of <A> tag?
   a. To insert an image b. To create a link
   c. To create a hyperlink d. To create a list
6. What is the output of given C program?

```
void main(){
    char str1[] = "FIRST"; char str2[20]; strcpy(str2,str1);
    printf("%s %s ",str1,str2); printf("%d", (str1!=str2));
    printf("%d", strcmp(str1,str2)); }
```

   a. FIRST FIRST 0 0 b. FIRST FIRST 1 1
   c. FIRST FIRST 1 0 d. FIRST FIRST 0 1
7. Where is a class derived in inheritance?
   a. Superclass b. Subclass
   c. Subsetclass d. Relativeclass

8. Which of these is the correct order of the SDLC?
   a. Analysis, Design, Coding, Testing, Implementation
   b. Analysis, Design, Testing, Implementation, Coding
   c. Implementation, Coding, Analysis, Design, Testing
   d. Design, Testing, Implementation, Coding, Analysis
9. Why is cloud computing popular nowadays?
   a. Cost-sharing and easily accessible
   b. As modern technology and costly
   c. Accessible and freely available
   d. Affordable to all

Answers									
1.	2.	3.	4.	5.	6.	7.	8.	9.	
a	c	a	d	c	c	a	a	a	

Group B

Short Answer Questions **[5×5=25]**

1. Explain 2NF and 3NF with examples.
   OR
   Demonstrate the basic DML statement with an example.
2. Write a function to add any two numbers in Javascript.
   OR
   Demonstrate the external CSS implemented in the web page.
3. Describe any five features of OOPs.
4. What are the different stages of software planning? Describe.
5. Define the concept of AI and IoT. (2+3)

Group C

Long Answer Questions **]2×8=16]**

6. How do you implement the Class C IP address in the local area network? Describe.
7. Write a program to enter ten integer numbers into an array, sort and display them in ascending order.
   OR
   Write a program to read the marks of any 5 students in a subject and count how many students are pass and fail.

SAMPLE MODEL QUESTIONS

1

Group A

Multiple Choice questions **(9×1=9)**

Time: 20 minutes

Tick the correct answer.

1. The object of MS-Access that is used to generate a hard copy of records.
   a. Query b. Table
   c. Form d. Report
2. A . . . key uniquely identifies a record.
   a. Primary b. Foreign
   c. Composite d. None
3. Which of the following is a remote login service?
   a. Video conference b. Telnet
   c. FTP d. All of the above
4. . . . returns today's date and time.
   a. date () method b. data () method
   c. image () method d. None of the above
5. . . . extracts a section of a string and returns a new string.
   a. set() method b. slice() method
   c. split() method d. None of the above
6. Choose a correct statement about C Language Functions.
   a. A function name cannot be same as a predefined C Keyword.
   b. A function name can start with an Underscore (_) or A to Z or a to z.
   c. Default return type of any function is an Integer.
   d. All the above.
7. In . . . paradigm, programs are written as a collection of classes and objects.
   a. Procedural paradigm
   b. Structural paradigm
   c. Object-Oriented paradigm
   d. None of the above
8. DFD stands for . . . .
   a. Data flow Diagram b. Data file diagram
   c. Data fixed diagram d. None of the above
9. . . . is a process of buying and selling or exchanging products, services, and information using electronic media.
   a. E-commerce b. payment
   c. shopping d. Traditional commerce

Answers									
1.	2.	3.	4.	5.	6.	7.	8.	9.	
d	a	b	a	b	d	c	a	a	

Group B

Short Answer Questions **[5×5=25]**

10. What is an E-R diagram? Explain the advantages of the E-R diagram in system design.
    OR
    What is data security? How it can be implemented?
11. Explain the different operators used in a PHP.
    OR
    Write a JavaScript to print the largest & smallest numbers among 10 elements of an array.
12. What is OOP? List the application of OOP.
13. What is system testing? Explain different types of testing techniques during the system development.
14. Classify e-Commerce according to the nature of the transaction.

Group C

Long Answer Questions **]2×8=16]**

15. What is networking? Distinguish between star topology and ring topology of networking principles with the help of a clean diagram.
16. What is a data file? Write down the syntax to create a new file.
    OR
    Write a C program to open a new file and read roll-no, name, address and phone number of students until the user says "no", after reading the data, write it to the file then display the content of the file.